

Harun Emrecan Karabağ

Game Developer | Unity | C# | Gameplay Systems

Türkiye | Open to Game Developer / Unity Developer roles | +90 507 049 3574 | harunemrecankarabag@gmail.com

Portfolio: [hekyetmisbes.github.io](https://github.com/hekyetmisbes) | Itch.io: hekyetmisbes.itch.io | GitHub: github.com/hekyetmisbes | LinkedIn: linkedin.com/in/harunemrecankarabag

Professional Summary

Game Developer and Computer Engineer focused on Unity, C#, gameplay systems, and playable game prototypes. Experienced in building complete gameplay loops, player movement, mission systems, progression, scoring, validation logic, UI/HUD flows, player feedback, Android build preparation, and published Windows playable builds.

Currently developing a stronger game engineering profile around gameplay systems architecture, movement and interaction polish, prototyping new game concepts, and design patterns for maintainable game code. Comfortable working across Unity 2D/3D projects, casual and puzzle mechanics, simulation gameplay, driving systems, and mobile-friendly player experiences.

Core Skills

Game Development: Unity 2D/3D, C#, gameplay programming, prototype development, casual games, puzzle games, platformers, simulation prototypes, playable build delivery

Gameplay Systems: player movement, interaction mechanics, scoring, progression, mission flow, quest systems, collectibles, validation logic, win/loss conditions, player feedback, UI/HUD systems

Unity Workflow: Scene Management, Prefab workflows, Unity UI, Input System, Cinemachine, ScriptableObject-oriented data structures, build testing, Android build pipeline, Windows builds

Architecture & Code Quality: OOP, SOLID principles, clean code, modular systems, separation of concerns, reusable components, maintainable gameplay architecture, design-pattern learning

Performance & Debugging: debugging, optimization awareness, object lifecycle management, pooling awareness, memory-conscious implementation, CPU/GPU performance fundamentals, profiling-oriented mindset

Tools & Collaboration: Git, GitHub, GitLab, documentation, code review readiness, issue-based workflows, technical communication, AI-assisted debugging and refactoring

Programming: C#, C++, Java, Kotlin, ASP.NET Core basics, EF Core basics, Firebase, SQLite concepts, JSON/data handling concepts

Additional Engines & Tools: Unreal Engine fundamentals, Figma for UI concepts, React basics

Selected Game Projects

Delivery Driver - Open-City Delivery Simulation Prototype

[GitHub](#) | [Playable Build](#)

Stylized open-city driving and delivery prototype focused on route planning, mission-based delivery flow, vehicle progression, traffic behavior, minimap navigation, runtime UI, and performance-aware systems.

Technologies: Unity 3D, C#, Input System, Cinemachine, SQLite, UI/HUD systems, NPC traffic AI, pooling, Git

- Built and evolved a Unity 3D driving prototype combining car control, delivery missions, route guidance, minimap markers, speedometer HUD, settings UI, and player progression.
- Implemented gameplay systems around runtime mission generation, pickup/delivery flow, money/XP/level progression, achievements, cargo damage, quest outcomes, and player statistics.

- Worked on simulation-oriented systems such as NPC traffic behavior, road graph concepts, obstacle avoidance, update throttling, pooling, world chunk management, and performance optimization structures.
- Published a downloadable Windows build on Itch.io and maintained the project through iterative updates and development logs.

The Platform - 2D Platformer Game

[GitHub](#) | [Playable Build](#)

Released 2D platformer where players collect 30 stars, overcome obstacles, complete levels, and interact with a simple cyberpunk-themed gameplay loop.

Technologies: Unity 2D, C#, platformer mechanics, level flow, collectibles, ShaderLab/HLSL, Git

- Developed 2D movement, obstacle interaction, level progression, collectible-driven goals, and completion conditions for a playable platformer experience.
- Designed a simple but readable gameplay loop around collecting stars, navigating levels, avoiding hazards, and improving completion performance.
- Published a Windows build on Itch.io, including install instructions and a downloadable executable package.

Flag Quiz - Android Mobile Quiz Game

[GitHub](#)

Mobile quiz game where players identify country flags through a simple, accessible, and score-driven gameplay loop.

Technologies: Unity 2D, C#, Firebase Realtime Database, Android, Unity UI, Figma, Git

- Built a mobile-friendly quiz loop with random flag presentation, answer validation, scoring, progress flow, and end-of-round feedback.
- Used Firebase Realtime Database concepts to manage flag names and country-code data, supporting dynamic content handling.
- Designed UI concepts with Figma and prepared Android-focused gameplay interactions with responsive player input.

Movidle Game - Movie Guessing Game

[GitHub](#)

Movie guessing game based on IMDB Top 250 data, comparing player guesses by movie attributes such as genre, year, director, country, and star.

Technologies: Java, game logic, validation systems, CSV/data handling, UI flow

- Developed a guessing-game logic where the system randomly selects a target movie and compares player predictions through multiple movie attributes.
- Implemented limited-attempt gameplay with feedback-driven comparison, success/failure states, and data-backed movie selection.
- Used this project as an early game-logic foundation before later strengthening Unity/C# gameplay development skills.

Professional Experience

Software Engineer Intern

Jul 2025 – Aug 2025

[Erciyes University IT Department](#)

Kayseri, Türkiye

- Developed structured and maintainable software components using ASP.NET Core 8 and EF Core, strengthening clean code, debugging, feature implementation, and software engineering discipline.
- Worked with data models, service logic, documentation, and iterative improvements to support stability, readability, and maintainability in an existing codebase.
- Used version control and integration-oriented development practices, building habits relevant to peer review, collaborative engineering, technical documentation, and end-to-end feature ownership.

Research Intern - 2247-C STAR

Feb 2024 – Jul 2024

[TÜBİTAK](#)

Advisor: Asst. Prof. Saltuk Buğra Selçuklu

- Conducted research on agent-based simulation and optimization, strengthening analytical thinking, experimentation, and structured problem-solving.

- Worked with simulation-oriented concepts relevant to real-time logic, decision flows, system behavior, and performance-conscious interactive software design.
- Applied an analytical approach to complex system behavior, supporting faster diagnosis of technical and design problems in gameplay and software systems.

Education

Hacettepe University

2026 – Present

M.Sc. in Computer Animation and Game Technologies

Expected Graduation: Jan 2027

Erciyes University

2020 – 2025

B.Sc. in Computer Engineering

GPA: 3.00 / 4.00

Certificates

- More C# Programming and Unity - Coursera: [Certificate](#)
- Introduction to C# Programming and Unity - Coursera: [Certificate](#)
- Hypercasual Fundamentals - BTK Akademi: [Certificate](#)
- Version Control: Git and GitHub - BTK Akademi: [Certificate](#)

Languages

Turkish: Native | **English:** B2+ written and spoken